

Interferometric scattering microscopy reveals microsecond nanoscopic protein motion on a live cell membrane

Richard W. Taylor^{1,2}, Reza Gholami Mahmoodabadi^{1,2}, Verena Rauschenberger³, Andreas Giessel³, Alexandra Schambony^{1,3} and Vahid Sandoghdar^{1,2,4*}

Many of the biological functions of a cell are dictated by the intricate motion of proteins within its membrane over a spatial range of nanometres to tens of micrometres and time intervals of microseconds to minutes. This rich parameter space is not accessible by fluorescence microscopy, but it is within reach of interferometric scattering (iSCAT) particle tracking. However, as iSCAT is sensitive even to single unlabelled proteins, it is often accompanied by a large speckle-like background, which poses a substantial challenge for its application to cellular imaging. Here, we employ a new image processing approach to overcome this difficulty and demonstrate tracking of transmembrane epidermal growth factor receptors with nanometre precision in all three dimensions at up to microsecond speeds and for durations of tens of minutes. We provide examples of nanoscale motion and confinement in ubiquitous processes such as diffusion in the plasma membrane, transport on filopodia and rotational motion during endocytosis.

Recent steady and rapid progress in fluorescence microscopy has brought us closer to monitoring cellular events at nanometre-scale spatial resolution, however, there still remains a great deal to accomplish¹. A major challenge is that the finite emission rate of a fluorescent source such as a dye molecule or a semiconductor quantum dot precludes high localization precision within microseconds, because too few photons are emitted in very small windows of time. Fluorescent probes also succumb to photodegradation, which restricts the duration of observation. Rayleigh scattering from noble metal nanoparticles has been detected through dark-field, differential interference contrast or bright-field microscopy^{2–5}, overcoming these limitations. A central difficulty in scattering-based microscopy is that the contribution from a nanoscopic probe competes against background scattering and a low signal-to-noise ratio (SNR). As a result, such efforts have been limited in positional uncertainty to several tens of nanometres in high-speed tracking experiments⁶, unless large probes were used⁵.

In this work, we employ interferometric scattering (iSCAT) microscopy^{7–9} to track proteins in live cell membranes, which presents a substantial spatial and temporal leap over fluorescence-based developments in visualizing the interaction of a probe with the cell, and the association between diffusional dynamics and local topology. We use gold nanoparticles (GNPs) to label epidermal growth factor receptors (EGFRs) in HeLa cells. An EGFR is a single-pass, type I transmembrane protein and a member of the receptor tyrosine kinase family, which senses and responds to extracellular signals and whose aberrant signalling has been linked to a variety of cancers^{10,11}. In this work, we provide examples of sub-diffusion and nanoscopic confinement in the three-dimensional (3D) motion of a protein at a very high temporal resolution and for a long duration. Furthermore, we show that a GNP-labelled

protein can be used as a ‘nano-rover’ to map the nanoscopic topology of cellular features such as membrane terrains, filopodia or clathrin structures.

3D iSCAT microscopy

The principle of iSCAT microscopy is that light scattered from a nanoparticle of interest is interfered with a brighter reference field, usually one reflected from the sample substrate⁷ (further details in Supplementary Section 1). The high SNR and sensitivity of iSCAT have allowed fast detection and tracking of nano-objects as small as single unlabelled proteins, but this has been mostly achieved in model synthetic systems with minimal background scattering, such as clean substrates^{12–14}, supported bilayers^{15–17} or giant unilamellar vesicles¹⁸. When iSCAT microscopy is applied to cells¹⁹, the local roughness of the plasma membrane and the nanoscopic heterogeneity of the cellular corpus give rise to a dynamic, speckle-like background, which substantially reduces its performance.

In this work, each GNP was functionalized with epidermal growth factor (EGF) proteins, and we verified that the probes stimulate the EGFR (Supplementary Section 2). Details of iSCAT microscopy, probe functionalization and cell culturing are provided in the Methods. As illustrated in Fig. 1a, EGF-GNP probes were introduced to the sample chamber of the microscope by a micropipette to label endogenous EGFR on HeLa cells. Recent investigations²⁰ suggest that although the probe size can influence rates of lipid diffusion in synthetic membranes, it does not affect the mode of diffusion. In live cells, another recent study reported²¹ negligible effects of molecular crowding for particles equal to or smaller than 50 nm. We verified this in two concrete cases by comparing data recorded with GNPs with 48 nm and 20 nm diameters. Furthermore, our fluorescence and biochemical studies suggest that the EGF-coated GNP activates EGFR signalling in the same way as

¹Max-Planck-Zentrum für Physik und Medizin, Erlangen, Germany. ²Max Planck Institute for the Science of Light, Erlangen, Germany. ³Department of Biology, Friedrich Alexander University Erlangen-Nuremberg, Erlangen, Germany. ⁴Department of Physics, Friedrich Alexander University Erlangen-Nuremberg, Erlangen, Germany. *e-mail: vahid.sandoghdar@mpl.mpg.de

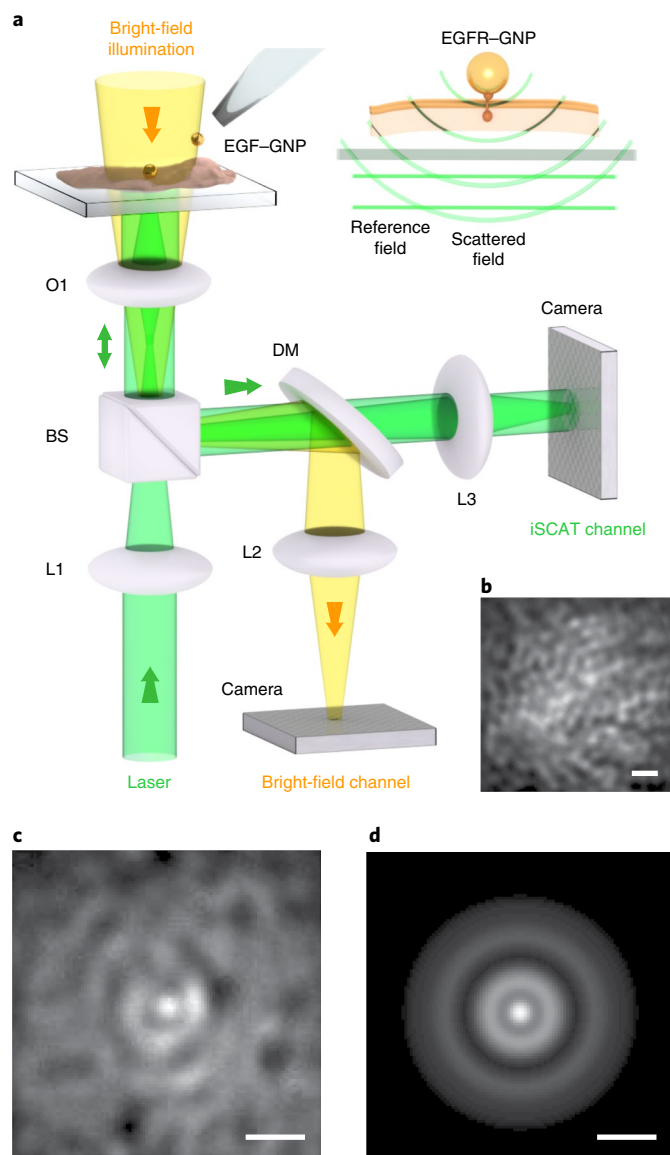


Fig. 1 | iSCAT microscopy on live cells. **a**, Experimental arrangement of the iSCAT microscope for live-cell imaging. Cells are plated in a glass-bottomed dish under Leibowitz medium. A micropipette delivers the EGF-GNP probes directly onto the cell culture, where they specifically target the EGFR protein in the cell membrane. The bright-field illumination from above assists in inspecting the culture but is not required for iSCAT imaging. L1–L3, lenses; O1, $\times 100$ objective; BS, 90:10 beamsplitter; DM, 590 nm short-pass dichroic mirror. iSCAT imaging was performed with illumination intensities of $1\text{--}8\text{ kW cm}^{-2}$, which are known to be viable for HeLa at our wavelength⁵⁰. Inset, wavefronts of the fields contributing to the iSCAT signal. **b**, A section of the membrane of the HeLa cell before labelling, viewed via reflection iSCAT. **c**, iSCAT image of the cell membrane including a bound EGF-GNP probe. **d**, The PSF extracted from **c**. Scale bars in **b–d** are $1\text{ }\mu\text{m}$.

free EGF, verifying that the label does not hinder biological function (see Supplementary Sections 2 and 3).

Figure 1b shows an iSCAT image of the upper surface of a HeLa cell before EGF-GNPs were added, revealing speckle-like contrast variations of up to 50% and spatial features as small as the diffraction limit. In Fig. 1c, we present a snapshot of an EGF-GNP bound to the plasma membrane, yielding a positive contrast (see Supplementary Video 1). The challenge with live-cell iSCAT microscopy is accounting for this dynamic background and extracting the point-spread

function (PSF) of the nanoparticle under study to reconstruct its 3D trajectory at high speed. Recent efforts in applying iSCAT to cellular imaging have fitted the PSF with a Gaussian profile and have tackled fluctuating backgrounds with limited success through temporal median averaging^{22,23}. Here, we implement a new algorithm to extract the full iSCAT-PSF directly from each frame. Figure 1d displays the PSF that is obtained from the image in Fig. 1c. The series of concentric rings of variable contrast is caused by the interference between the plane and spherical waves incident on the camera (see Fig. 1a). We determined the lateral displacements of the GNP by localizing its PSF through pairwise frame cross-correlation. Doing so circumvents the difficulties associated with fitting Gaussian profiles to the complex iSCAT-PSF (see the Methods and Supplementary Sections 4 and 5).

The interferometric nature of the iSCAT signal also leads to contrast modulations as the scatterer moves in the axial direction. 3D iSCAT tracking has been demonstrated by monitoring the brightness of the central PSF spot^{18,24}. However, the ambiguity caused by the periodicity of the PSF brightness limits its applicability to axial displacements on the order of 100 nm. To remedy this issue, we exploit the information encoded in the amplitudes and radial positions of the PSF rings, which also change along the optical axis. We take advantage of the fact that these features do not undergo symmetric oscillations as a function of axial displacements if the focus is placed at a distance of several micrometres above the glass interface; that is, where the GNP lies on the cell membrane (see Supplementary Section 5). Together with the large SNR in our measurements, our new image processing procedures result in a localization precision of $\sigma = 2\text{--}6\text{ nm}$ for both the lateral and axial directions, taking us beyond conventional 2D trajectories that can be commonly found in the literature²⁵. A comparison of our new approach against previous works, and further details, are provided in Supplementary Sections 4–6.

Results

The spatiotemporal dynamics of protein function in the membrane, as well as the protein's uptake and trafficking, are expected to be strongly influenced by heterogeneous interactions with various cellular components such as actin filaments, clathrin structures, other membrane proteins and the extracellular matrix²⁶. A lack of means to directly visualize these features at high spatial and temporal resolution has left much of their details a matter of debate. We now discuss numerous trajectories that reveal a wealth of dynamic 3D heterogeneities and shed new light on protein motion.

Quantitative study of subdiffusion in the plasma membrane. We begin by considering a 2D example of the EGFR journey on the plasma membrane of a living HeLa cell. Figure 2a presents a trajectory that results from a video containing $N = 75,000$ frames over 3.75 s recorded at 20,000 frames per second (fps). The high SNR yields $\sigma = 6\text{ nm}$ lateral spatial precision within the frame exposure time of $17.5\text{ }\mu\text{s}$. A visual inspection of the trajectory, which is temporally colour coded, indicates strong heterogeneity.

To analyse the diffusion behaviour in a quantitative fashion, we first compute the mean square displacement (MSD) for the whole trajectory by employing the conventional formula of $\text{MSD} = 4D\tau + 2\sigma^2$, where D is the diffusion constant, τ is the time lag and the term $2\sigma^2$ represents the mean localization error in x and y (ref. 27). The blue curve in Fig. 2b presents the MSD, which displays a systematic and non-trivial deviation from the expected behaviour of normal diffusion (illustrated by the black straight line). The observed MSD shows a substantial tendency towards subdiffusion and, hence, the existence of binding or hindering interactions²⁸. We point out that the localization precision in most published works is only determined categorically for the microscope (for example refs. 3,22), but in our experiments it was assessed for each frame. We

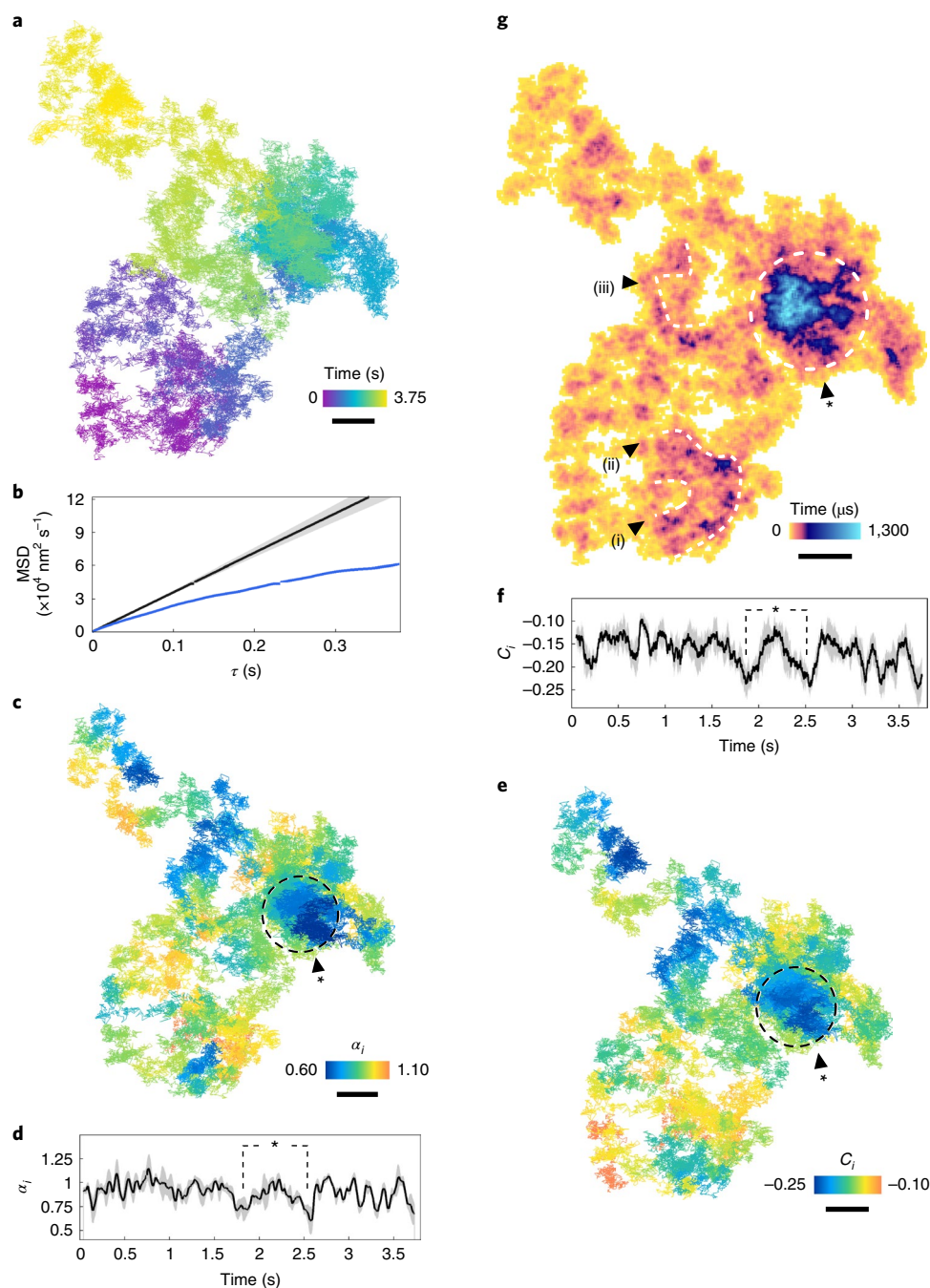


Fig. 2 | Diffusion on the plasma membrane. **a**, A lateral diffusional trajectory (17.5 μ s exposure time, see colour scale for chronology). **b**, MSD versus τ . The blue curve shows the MSD of **a**. The black curve is simulated normal diffusion ($\alpha=1$), with the grey envelope indicating the uncertainty. **c**, The diffusional exponent of rolling windows (colour scale) over the trajectory. Regions of subdiffusion ($\alpha<1$) are indicated by darker shades. **d**, α_i through time. The grey shading represents a mean uncertainty of $7 \pm 4\%$, corresponding to a 95% confidence interval for a window of 100 ms (1,000 frames) and $\tau=250 \mu$ s. The points marked with the asterisk correspond to the circle in **c**. **e**, The step-direction C_i for rolling windows along the trajectory. **f**, The step-direction C_i plotted through time, with the shading denoting uncertainty. **g**, ATOM occupation plot with residency time (colour scale). The bin size corresponds to the localization error. Noteworthy regions of extended occupation, marked as loops and whirls (i)–(iii), are indicative of persistent nanoscopic structures. The enclosed region represents a dense patch of notable subdiffusion. Scale bars, 100 nm.

also emphasize that the inclusion of the positional uncertainty (σ) is important for a correct assessment of the MSD²⁷, especially in high-speed imaging where the MSD becomes comparable to the localization precision.

The occurrence of transient nanoscale phenomena such as lipid rafts or protein clustering has been a topic of great interest regarding diffusion in the plasma membrane^{26,29}. Efforts to identify local

confinements typically fall within two approaches, both based on assumptions. One strategy builds on a priori knowledge of the effective diffusion constant and infers confinement when displacement falls short of the expected travel³⁰. An alternate approach considers a specific geometry and fits the measured MSD to a model derived for that scenario; for example, hopping across semi-permeable barriers of square confinements³¹ (see examples in ref. ³²). We do not

need to make any assumptions about the nature of the diffusion or its geographic landscape because the very large number of iSCAT frames and the localization precision in each frame make it possible to segment each trajectory into small windows while ensuring robust statistics.

We examine local deviations from normal diffusion by considering:

$$\text{MSD}_i = 4D_i\tau^{\alpha_i} + 2\sigma_i^2 \quad (1)$$

for the i th rolling temporal window. This formulation provides a convenient measure for the nature of diffusion: α is unity for normal diffusion, $\alpha < 1$ indicates subdiffusion and $\alpha > 1$ is a sign of superdiffusion³³. In Fig. 2c, we map α_i in a colour-coded fashion by indicating varying degrees of middling and strong confinement. An example of the latter is the region marked with the dashed circle, which will be explored further below. To facilitate a quantitative insight into the overall trajectory, we also plot α_i as a function of time in Fig. 2d, whereby the grey regions depict the uncertainty (see Supplementary Section 7 for the contribution of errors to α_i). The regions where α_i clearly deviates from unity seem to be locally confined to a few tens of nanometres and within time intervals of tens to hundreds of milliseconds. An example corresponds to the region marked with an asterisk in Fig. 2c where $\alpha_i < 0.75$. We note that our findings do not depend on the length of the rolling windows.

Another approach for gauging the occurrence of diffusive barriers and confinements is to consider the degree of directional correlation (C_i) between two vectorial steps across a time lag³⁴. A truly random process would possess no directional correlative memory (median $C_i = 0$), whereas negative or positive values would indicate reflections or forward push, respectively (see Supplementary Section 7). Figure 2e shows the 2D map of C_i , and in Fig. 2f we plot the correlation coefficients over sliding windows of 500 frames with the same temporal lag that was used to determine α_i . We find local variations around a baseline of $C = -0.15$, whereas comparison of Fig. 2e with Fig. 2c, and Fig. 2f with Fig. 2d, confirms a correspondence between the areas of pronounced subdiffusion ($\alpha_i < 1$) and strong backlash ($C_i < 1$).

As a third way of discovering heterogeneities, we assessed the popularity of each trajectory pixel in space. Here, we introduce the concept of an accumulated temporal occupancy map (ATOM), wherein we divide the lateral plane of the trajectory into nanometre-sized bins as small as the localization precision and count the occurrence of the particle in each bin. The areas of extended residency in Fig. 2g identify persistent nanoscopic structures arranged in loops and whirls (see labels (i)–(iii)) as well as a patch extending over approximately 150 nm marked by an asterisk, whereby the latter corresponds well to the marking in Fig. 2c. We observed a minimal lifetime of 250 ms (5,000 frames) for this feature, which might portray a pre-endocytic step (we will return to this discussion). Overall, the observed inhomogeneities strongly suggest that protein diffusion is affected by cellular substructure.

Diffusion along a filopodium over minutes. An important advantage of iSCAT microscopy is its ability to record very long events. We now apply this feature together with the 3D imaging capability of iSCAT to follow EGFRs on a filopodium. Filopodia are rod-like cellular protrusions that contain bundles of actin filaments that are 100–300 nm in diameter and up to 10 μm in length. They act as sensors for mechanical stimuli, putative attachment sites, chemoattractants or repellents in the surroundings of the cell. Filopodia also provide cellular signalling sites that are enriched in growth factor receptors^{35,36}, and it has been shown that ligand binding and EGFR activation on filopodia occur preferentially at low EGF concentrations. Activation is then followed by association with the actin filaments and retrograde transport of EGFR to the cell body, where the

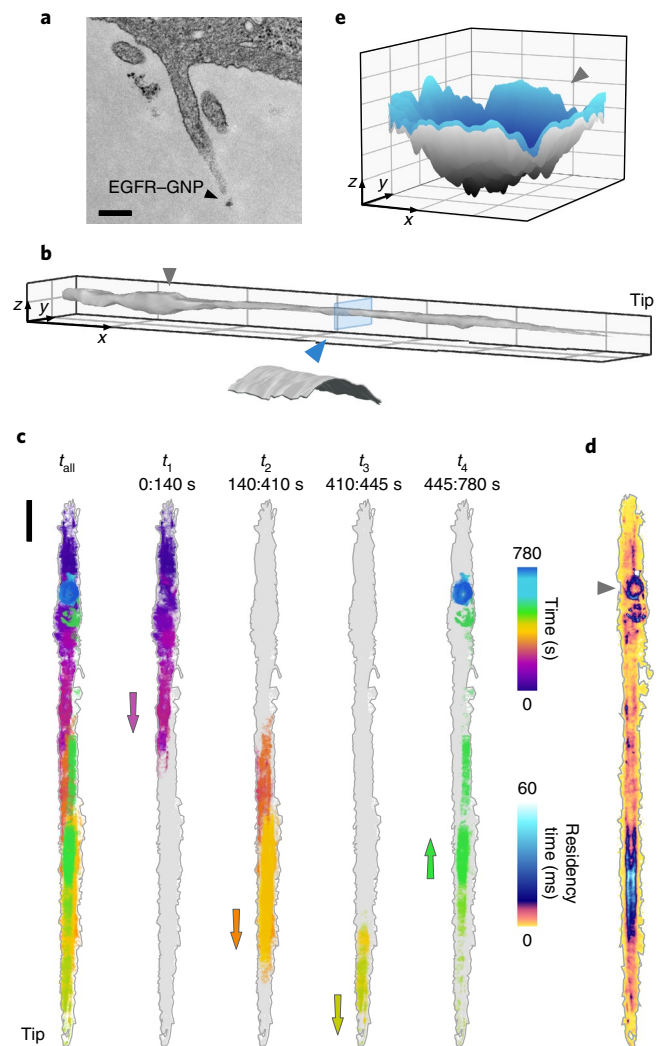


Fig. 3 | Mapping the journey of an EGFR. **a**, A TEM image of a filopodium including an EGFR-GNP. **b**, A filopodium surface reconstructed from 780,000 trajectory points with a localization error of $\sigma_{xy} = 2$ nm recorded at 1,000 fps. Inset, cross-sectional slice that depicts a cylindrical surface of diameter 150 nm after accounting for the size of the GNP. **c**, A raw 13 min trajectory (left) broken into four subsequent pieces that reveal the journey to and from the tip, with arrows marking direction of net motion. **d**, An ATOM plot of **c**, corrected for filopodium drift (see Supplementary Section 8). **e**, A surface interpolation plot from the final 80 s. The ring-like confinement in the final phase (marked with a triangle) is a 3D pit. The scale bars are 200 nm (**a**), 1 μm (x, y) and 200 nm (z) (**b**), 1,000 nm (**c**) and 100 nm (x, y) and 50 nm (z) (**e**).

receptors are endocytosed^{37,38}. We now show how iSCAT microscopy gives insight into the nanoscopic details of this process. To set the stage, in Fig. 3a we display a transmission electron microscope (TEM) snapshot of a typical filopodium from a HeLa cell, including an EGFR-GNP.

In Fig. 3, we present data obtained from an iSCAT video recorded over the course of 13 min (see Supplementary Video 2). First, we analyse the 3D trajectory of the EGFR-GNP; the very large number of trajectory points makes it possible to interpolate among them and create the topography of the filopodium surface displayed in Fig. 3b (see also Supplementary Section 8). The 3D locations of the recorded positions map to a cylinder-like shape with a diameter that, when corrected for the diameter of the GNP, amounts to about

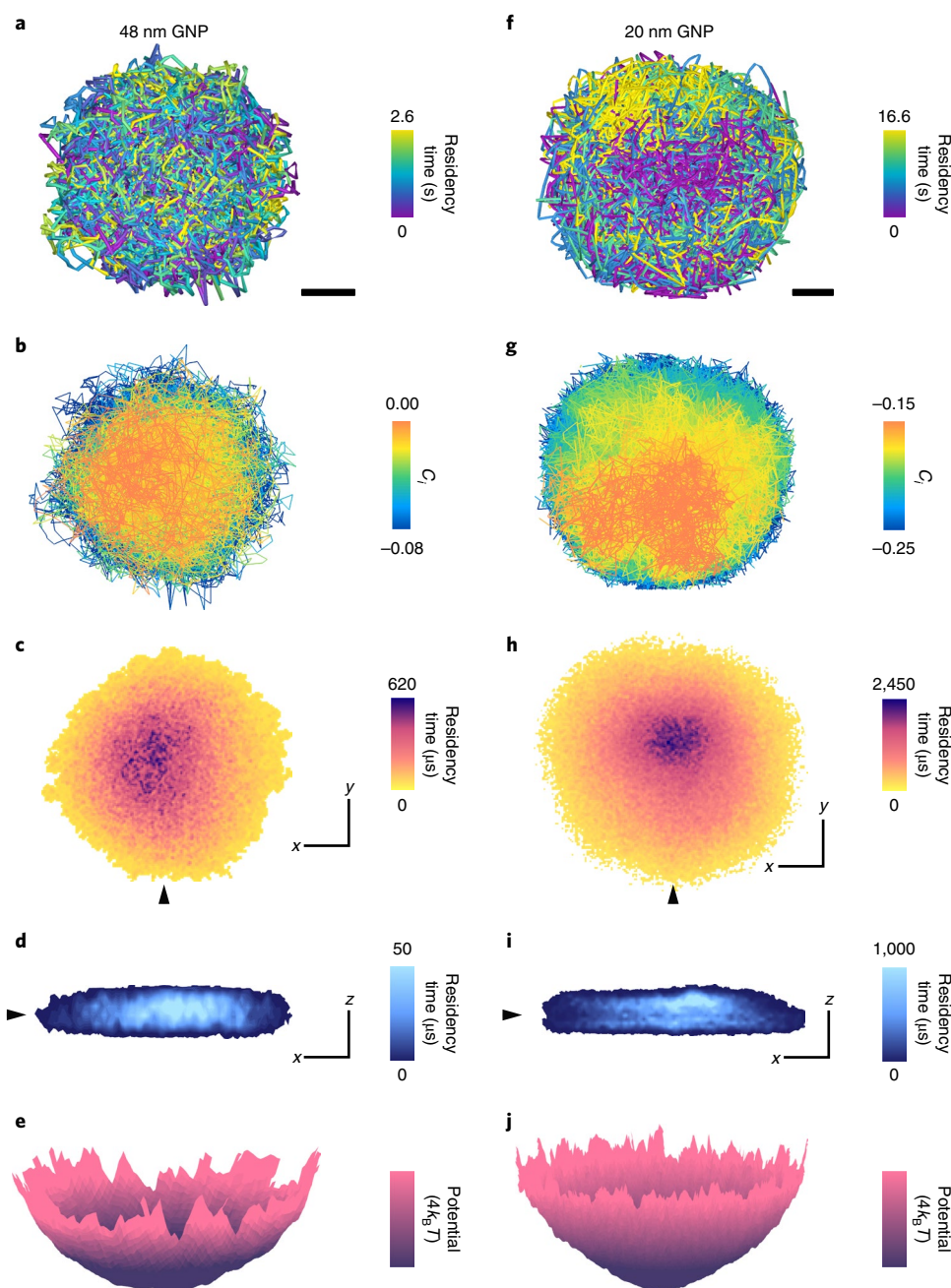


Fig. 4 | Confined diffusion recorded at 30,000 fps with 48 and 20 nm GNPs. **a**, A lateral trajectory of a 48 nm GNP probe. Scale bar, 100 nm. A lower temporal sampling of this confinement would have underestimated the extent of bounding⁴¹. **b**, C_1 of the trajectory (using a time lag of five frames), which shows partially hindered diffusion with a propensity for freer diffusion in the centre. **c**, An ATOM plot of **a**. **d**, A cut through the 3D-ATOM plot along the line of the black triangle in **c** shows that occupancy favours an innermost disk-like region. The axes denote 100 nm in both **c** and **d**. **e**, Conversion of the temporal 2D occupation from **c** into an effective potential energy distribution. **f–j**, Equivalent to **a–e**, but for a 20 nm GNP probe.

150 nm. The cut through the filopodium displayed in the inset clearly shows that the probe has explored the outer surface. This provides an example of a GNP serving as a nano-rover for mapping the topology of cellular structures, while its 3D coordinates are registered via iSCAT.

In Fig. 3c, we examine the full trajectory of the EGFR–GNP in more detail by visualizing sections from different time intervals. The colour-coded temporal evolution of the trajectory reveals a directionality accompanied by a longitudinal dithering behaviour. We find that the receptor initially travels towards the tip of the filopodium over a period of 445 s, which is followed by retrograde

trafficking back towards the cell body, where it is trapped in a confinement. This observation supports the literature reports of directed transport of receptor-bound EGF along filopodia³⁷.

To learn more about the phenomena at work, in Fig. 3d we plot the trajectory ATOM. We find that the occupancy of the filopodium tip region is short compared to the residence time along the shaft. The ATOM data also reveal extended residency in a circular patch and a ring-like structure towards the end of the trajectory (marked with a triangle). Interestingly, the 3D representation of this confinement in Fig. 3e indicates a pit-like topology, which would be consistent with a pre-endocytic step of membrane invagination.

A closer look at examples of confined diffusion and evidence for uptake. Another asset of iSCAT stems from the linearity of the scattering signal and lack of saturation, which limits the speed of fluorescence imaging. High-speed microscopy is not only necessary for obtaining high-resolution temporal information, but it also circumvents blurring effects in the spatial localization of nanoparticles³⁹. Figure 4a shows a bounded trajectory within a disk of 500 nm diameter over 2.6 s, an observation that is consistent with the long time confinements reported for EGFR^{3,40}.

The C_i and the 3D-ATOM in Fig. 4b–d provide further evidence for confinement in all dimensions. In particular, the lower C_i values on the outer part of the trajectory clearly represent an act of confinement, which is supported by a propensity for the protein to be found in the centre of the ATOM plot. We used the ATOM data to compute the normalized probability amplitude of spatial occupation ($P(x, y)$), and deduced an effective bounding potential according to $U = -k_B T \log(P(x, y))$ (ref. ⁴¹), where T is temperature and k_B is the Boltzmann constant. We find a concave functional form with a depth of $\sim 4k_B T$ (Fig. 4e), consistent with the results of ref. ⁴¹. The data in Fig. 4f–j present other examples of the same confinement behaviour recorded with a 20 nm GNP.

In Fig. 5a, we present another 3D trajectory recorded at a very fast rate of 66,000 fps with a short exposure time of 10 μ s and duration of 3.5 s, while maintaining a high spatial precision of $\sigma = 4$ nm (see Supplementary Section 5 for the error histogram). We note that determining the localization precision with a shorter exposure time and faster imaging can, in fact, be superior due to smaller changes in the background. To portray the axial variations of the particle coordinates in a quantitative fashion, in Fig. 5b we plot the height as a function of time with the same colour scale as in Fig. 5a. We see that the protein undergoes a substantial out-of-plane motion of up to 200 nm over 2 s. For comparison, we also plot the height variation of the membrane as a whole during the same period, which was deduced by analysing the iSCAT background.

Next, in Fig. 5c we present the ATOM plot of this trajectory, which, again, indicates a good deal of heterogeneity in residence times. A notable feature is the depressed valley region, where the protein spends more time. Figure 5d shows a magnification of this patch, which reveals circular features 25 nm in diameter that are arranged in a quasi-lattice. The overall size of the confinement and the dimensions of the underlying network of rings surveyed by the nano-rover are consistent with the features of a clathrin-coated lattice or pit⁴². It is known that the assembly of a clathrin coat on the inner surface of the plasma membrane results in the formation of clathrin-coated patches and can occur spontaneously or in a cargo-triggered manner. Once associated with a cargo protein, basket-like structures grow to form clathrin-coated pits (CCP), inducing membrane invagination. These CCPs are reported to cover about 1.5% of the surface of a HeLa cell, and are accompanied by flat clathrin lattices that can cover up to 8% of its inner membrane⁴³.

Epidermal growth factor receptor is expected to undergo clathrin-mediated endocytosis in HeLa cells when stimulated by low (picomolar) concentrations of EGF⁴⁴, a phenomenon that we confirmed through fluorescence confocal microscopy (see Supplementary Section 3). In Figs. 3–5, we showed that fast iSCAT microscopy can reveal intricate 3D features that hint towards endocytotic events. We now provide more evidence.

Figure 6a displays a conventional 2D map of a confined trajectory over a duration of 1 s, whereas its 3D smoothed map in Fig. 6b illustrates a bowl-like topology with a diameter of 150–200 nm (after correction for the size of the GNP) that lasts for tens of seconds, which is consistent with the literature knowledge on clathrin-mediated endocytosis^{45–47}. The fact that the GNP maps a curved surface suggests that the EGFR is not only engulfed by the pit, but also stays attached to and explores its inner surfaces. In Fig. 6c, we scrutinize the GNP trajectory by plotting its angular position as a

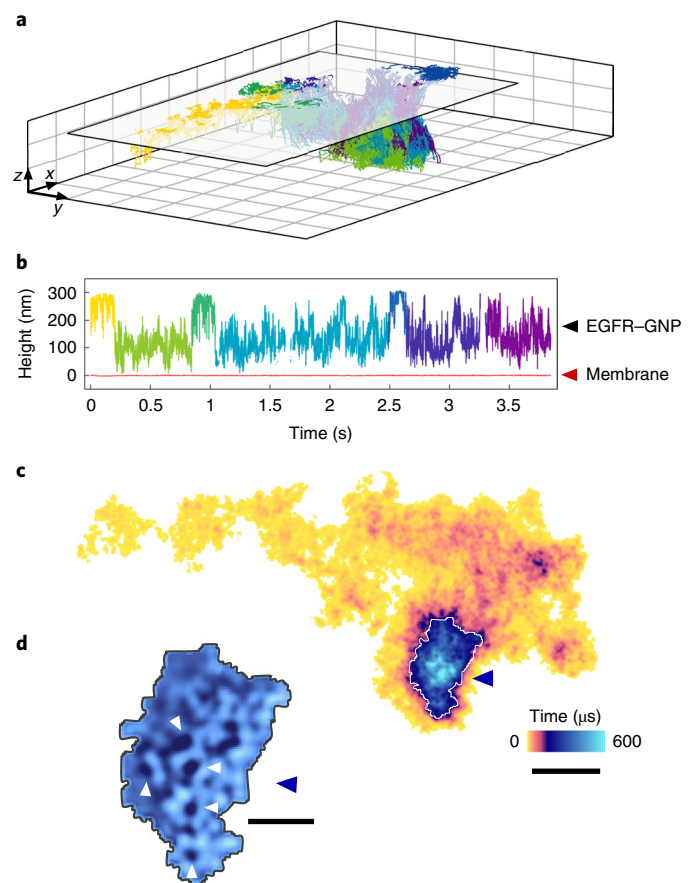


Fig. 5 | Ultra-high-speed 3D tracking at 66,000 fps. **a**, A 3D trajectory over 3.75 s with $\sigma_{xy} = 4$ nm. The initial surface of diffusion, marked with a white plane, sits above a depressed valley region. The scale bars are 200 nm (x, y) and 100 nm (z). The colour scale represents time according to the representation in **b**. **b**, The height of the probe as a function of time emphasizes substantial axial motion of the EGFR-GNP probe. The instantaneous background fluctuations of the cell membrane (red) corresponding to a time-averaged value of 0.5 nm show motion that is clearly uncorrelated with that of the EGFR-GNP probe. **c**, An ATOM plot. A patch of extended occupation (blue region, marked with blue triangle) is easily identified. The scale bar is 250 nm. **d**, On closer inspection after removal of the background, a network of 25 nm circular features (white triangles) becomes evident. The scale bar is 50 nm.

function of time portrayed in the radial direction. Interestingly, we observe a clear rotational behaviour. This evidence for a directed motion of the EGFR might point to the twisting action of dynamin, which is thought to be involved in closing CCPs⁴⁸. Figure 6d–f shows the equivalent data recorded from a 20 nm GNP. To verify that our EGFR-GNP entities are captured in CCPs, we also performed electron microscopy on our samples. Figure 6g presents an example of this, where EGFR-GNPs (blue triangles) are located in a CCP (white triangle) and a vesicle (yellow triangle).

We note that the finite GNP diameter does not seem to be prohibitive in our studies. This is not surprising given the much larger size of the CCP. In addition, considering that EGFR is expected to be engulfed in small lipid vesicles of size comparable to our GNPs after its uptake^{46,47}, the GNP size does not necessarily impose extra space constraints. Moreover, the data in Figs. 4 and 6 indicate that a reduction of the probe size from 48 nm to 20 nm does not lead to any notable change. The sizes of putative CCPs and internalized vesicles observed in iSCAT match those seen in our TEM images

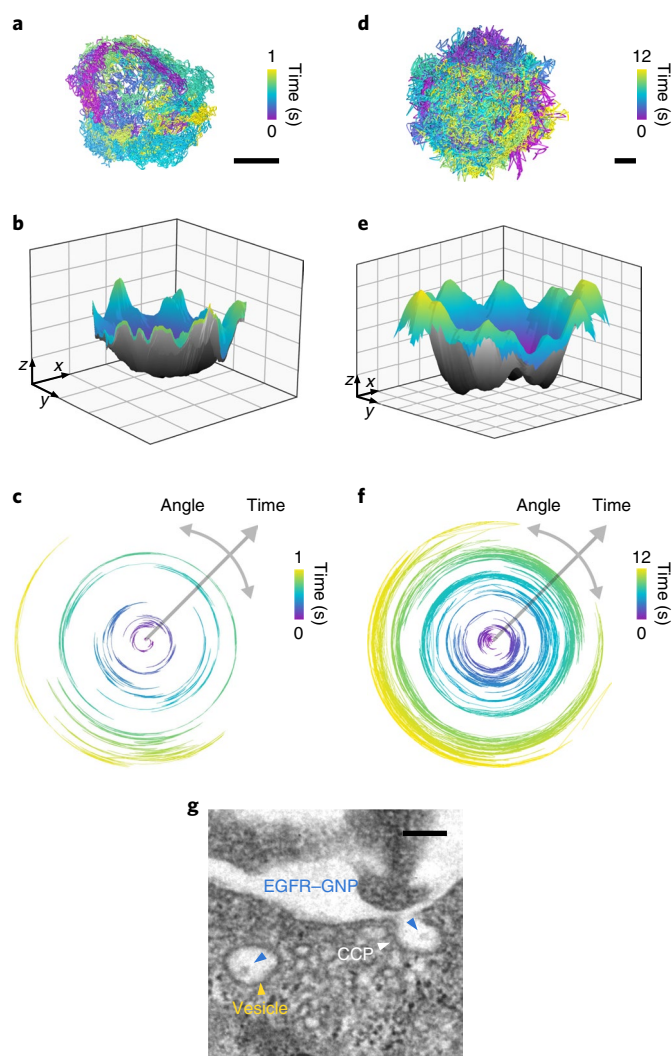


Fig. 6 | Confined diffusion within a pit. **a**, A lateral trajectory of a confined EGFR-GNP (diameter 48 nm) recorded at 20,000 fps with 17.5 μ s exposure time. Scale bar, 100 nm. **b**, The 3D projection of **a** maps to a bowl-like surface. The scale bars are 100 nm (x,y) and 20 nm (z). **c**, An angular trajectory of the GNP about the centre of rotation (the radius and colour represent time) indicated directed rotation. For clarity, only every 150th frame is plotted. **d-f**, Same as **a-c**, but with a 20 nm GNP probe recorded at 10,000 fps with an exposure time 50 μ s. Scale bar, 100 nm. **g**, An ultra-thin (50 nm) electron microscopy section of a HeLa cell that confines an EGFR-GNP probe (blue triangles) to a clathrin-coated pit (white triangle) and a vesicle (yellow triangle). Scale bar, 200 nm.

with and without EGFR-GNPs present, which indicates a lack of systematic artefacts (Supplementary Section 9).

Discussion and outlook

After several decades of work, and experimentation with a wide range of biochemical, analytical and imaging techniques, our knowledge of the cellular plasma membrane continues to be limited. In particular, it is still unclear how lipids, proteins and small molecules within the membrane interact with each other and with their neighbours, such as other proteins, the cytoskeleton or extracellular matrix. In the case of transmembrane proteins, different models and hypotheses have been put forth to explain the observed inhomogeneities in their diffusion and transport behaviour. In this work, we have presented intriguing data for which the combination

of very high spatial and temporal resolution provides new insight into subdiffusion, nanoscopic confinement and 3D contours of filopodia and clathrin structures. However, we have not attempted to verify or refute any of the existing hypotheses because drawing robust conclusions about complex biological processes requires dedicated endeavours that involve a series of independent control experiments. Our main focus has been to advance iSCAT microscopy for quantitative studies in cell biology.

In the future, we plan to combine iSCAT with in situ super-resolution fluorescence microscopy to correlate iSCAT trajectories of proteins and viruses with the structures of other entities such as actin filaments, microtubuli, clathrin or dynamin. For example, it would be interesting to investigate hypotheses such as hopping or corralling by the actin mesh²⁶, as well as temporal and spatial details of endocytosis⁴². More advanced image analysis methods should make it possible to track GNPs smaller than 20 nm, but a particularly promising application of our methodology will be in understanding the life cycle of viruses, which provide a large iSCAT signal without the need for an external label^{9,22,49}.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at <https://doi.org/10.1038/s41566-019-0414-6>.

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Author contributions

R.W.T. made iSCAT measurements and performed data analysis. R.G.M. performed the quantitative analysis of iSCAT images and trajectories. V.R. prepared TEM samples and carried out immunofluorescence and western blot experiments. A.G. performed electron microscopy. A.S. supervised biological preparation and data interpretation. V.S. conceived and supervised the project. V.S., R.W.T. and A.S. wrote the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to V.S.

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Methods

Synthesis of EGF–GNP probes. Gold nanoparticles with diameters of 48 nm, functionalized with monoclonal anti-biotin (British Biocell International), and GNPs with diameters of 20 nm, functionalized with streptavidin (British Biocell International), were conjugated at a molar ratio of 1:1 with biotinylated EGF (ThermoFisher) at a concentration of 0.66 nM. PBS, supplemented with 1% w/v biotin-free Bovine serum albumin (PBS–BSA), was used as a buffer. The solution containing EGF–GNP was agitated for several hours at 37°C to assist conjugation, purified through a NAP-5 column (GE Healthcare) and diluted to a final concentration of 0.1 nM.

Cell culture. HeLa cells were grown in DMEM (Gibco Invitrogen) supplemented with 10% fetal calf serum (FCS, Life Technologies) in a humidified atmosphere at 37°C and 5% CO₂. For measurement, the cells were plated onto a glass-bottomed sample dish (Ibidi GmbH) and grown to 70% confluency. Before measurement, each dish was rinsed twice with warmed PBS–BSA solution, serum starved for several hours and rinsed again with warmed PBS–BSA. When placed on the microscope, a buffer of Leibovitz's L-15 Medium (1.5 ml Gibco, Invitrogen) was added and the culture was held at 37°C by a home-built objective heater. A pipette was used to deliver 10 µl of the EGF–GNP to a local region of the culture, with observation beginning immediately thereafter.

Immunofluorescence. HeLa cells were seeded on fibronectin-coated cover-slips inserted into 24-well plates (Sarstedt) to a confluency of 1×10^5 cells per well. After seeding for 24 h, the cells were serum starved (DMEM, 0.1% FCS) for 16 h. The plates were chilled on ice for 1 h and then stimulated for 4 min at 37°C with either the control medium or 1 pM EGF–GNP with additional pre-warmed DMEM (with 0.1% FCS). After stimulation, the samples were washed twice with ice-cold PBS (Gibco, Life Technologies) and fixed with 4% paraformaldehyde for 15 min. The cells were then permeabilized with 0.1% TritonX-100 (Carl Roth) and blocked with 10% horse serum and PBS supplemented with 0.1% Tween 20. Epidermal growth factor receptor was detected using a specific primary antibody (rat anti-EGFR; abcam ICR10, 1:1,000) and a secondary anti-rat-Cy3 antibody (Jackson Immuno Research, 1:400). Clathrin or caveolin were visualized using specific antibodies: rabbit anti-clathrin (Proteintech, 1:200) or rabbit anti-caveolin (Proteintech, 1:500) and a secondary anti-rabbit-Alexa488 antibody (ThermoFisher, 1:1,000). Nuclei were stained with 4',6-diamidino-2-phenylindole (DAPI). The samples were mounted in Mowiol and imaged with a confocal laser scanning microscope (Leica SP5).

Preparation of cell-lysates and western blotting. HeLa cells were seeded at a confluency of 3×10^5 cells per well into a 6-well plate (Sarstedt), and were then cultured for 24 h followed by overnight incubation in starvation medium (DMEM, 0.1% FCS). Stimulation was performed using 10 pM or 100 pM for EGF, EGF–GNP and control GNPs for 2 min. Cells were lysed in 250 µl of 20 mM 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES, pH 7.4), 150 mM NaCl and 0.5% NP-40 supplemented with protease inhibitor and phosphatase inhibitor cocktails (Roche) at 4°C. Lysates were cleared by centrifugation for 10 min at 4°C and 16,000 g. The total amount of protein was determined by using a bicinchoninic acid assay (Applichem). Equal amounts of protein per sample were mixed with sodium dodecyl sulfate sample buffer and denatured for 5 min at 95°C. The proteins were separated by gel-electrophoresis (10% polyacrylamide gel) and

transferred onto a polyvinylidene fluoride membrane (Roche) using a semi-dry blotting system (BioRad). The primary antibodies used were rabbit anti-EGFR (Cell Signaling Technologies (D38B1), 1:1,000), mouse anti-GAPDH (Proteintech, 1:2,000), rabbit anti-ERK1 and ERK2 (Cell Signalling Technologies, 1:1,000) and rabbit anti-pERK (Promega, 1:2,000). The secondary antibodies used were anti-mouse-alkaline phosphatase and anti-rabbit-alkaline phosphatase (Cell Signalling Technologies, 1:3,000).

Electron microscopy. HeLa cells were seeded at a confluency of 5×10^5 cells per well into 4-well-lumox chamber slides (x-well-Lumox, Sarstedt). Stimulation was performed as described for immunofluorescence using 1 pM EGF–GNP. After stimulation, the cells were rinsed twice with ice-cold PBS and fixed with 25% glutaraldehyde and 4% paraformaldehyde in 0.1 M cacodylate buffer (pH 7.4) for 15 min at 37°C. The cells were washed once with PBS and fixed again with 4% osmium tetroxide in 0.1 M cacodylate buffer (pH 7.4) for 1 h in the dark. The samples were dehydrated by an ascending acetone series and embedded in an Epon–acetone 100% mix (1:1) overnight at 60°C. The samples were embedded in pure Epon on the next day. Ultra-thin sections of 50 nm were cut by an ultra-microtome (Ultra cut E, Reichert & Jung). Thin sections were mounted on a copper grid and counter stained with a uranyl acetate–lead citrate mix. The samples were imaged on a transmission electron microscope (Zeiss).

PSF extraction and 3D localization. A typical iSCAT image in our work contains contributions from scattering by the particle and the cell. To extract the PSFs of the probe in different frames of a recorded video, we first fit a 2D Gaussian function to the image of the GNP in one of the video frames that possessed a sufficiently large positive contrast to enable precise determination of the centroid position. Next, we computed the radial median intensity about this centroid and used this information to reconstruct the PSF for that frame. This approach is made possible by the strong circular symmetry of the PSF against the asymmetric fluctuations of the speckle background. The PSF (PSF_i) extracted from a given video frame, V_i , is then cross-correlated with the next time-subsequent frame, V_{i+1} , to determine the probe location in the latter. Here, we computed a 2D intensity cross-correlation following $O(u, v) = \sum_{x, y} \text{PSF}_i(x, y) V_{i+1}(x-u, y-v)$. Next, a Gaussian function is fitted to the maxima in the resultant cross-correlation product O , and the uncertainty in the lateral position of the localized probe is computed from this fit (see Supplementary Sections 4 and 5). This approach is repeated throughout the entire video and is robust against changes in PSF contrast polarity. This algorithm relies on our fast imaging frame rate, which avoids large jumps in the trajectory. The concentric ring pattern of the PSF enables determination of the axial position of the probe.

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Data availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Code availability

Algorithms used in this study are available from the corresponding author on reasonable request.

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Data collection

All iSCAT videos were collected using either proprietary software (Phantom, VisionResearch) or custom written code in LabView (National Instruments). TEMs were captured using Gatan Microscopy Suite Version 2.3.1. Confocal Fluorescence images were captured with Leica LAS AF software version 2.7.3.9723.

Data analysis

All analysis was performed using custom written code in MATLAB (MathWorks). Immunofluorescence images were processed using OMERO and Image J software, Western Blots and TEM images were contrast adjusted and cropped using Adobe Photoshop 6 and grouped and labeled using Adobe Illustrator 5.

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Antibodies

Antibodies used	anti-clathrin-HC, proteintech, 26523-1-AP; anti-caveolin 1, proteintech, 16447-1-AP; anti-EGFR, abcam, ab231, clone ICR10; anti-EGFR, cell signaling technology, 4276, clone D38B1; anti-ERK, cell signaling technology, 4695, clone 137F5; anti-phospho ERK, Promega, V8031; anti-GAPDH, proteintech, 60004-1-Ig, clone 1E6D9; anti-mouse-AP, cell signaling technology, 7056; anti-rabbit-AP, cell signaling technology, 7054; anti-rabbit-Alexa 488, Thermo Fisher, A21206; anti-rat-Cy3, Jackson Immuno Research, 712-165-153
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